Assignment 4 - Report

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1 About NHANES dataset

Throughout our past assignments, we have regularly used a cleaned_NHANES dataset. This dataset was selected from *The National Health and Nutrition Examination Survey* (NHANES) data. It includes the blood pressure data and demographic data collected during three NHANES survey cycles:

- 2013-2014: (Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS) 2014)
- 2015-2016: (Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS) 2016)
- 2017-2018: (Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS) 2018)

This dataset has been cleaned and is provided as "cleaned_NHANES.csv". It contains 29,400 observations and has 21 different variables (some in character and others in integers).

Below, we will follow through a brief description of a couple of variables from the dataset, presented into two tables. The required packages and data are loaded and imported as such:

1.1 Load packages and data import

```
library(pacman)
pacman::p_load(rio, here, dplyr, gtsummary,
               forcats, kableExtra)

# Data import
cleaned_NHANES <- import(here("Data", "cleaned_NHANES.csv"))
```

2 Data description

As mentioned previously about the selected data from NHANES, we worked on data from 3 survey cycles. We will therefore be comparing between survey cycles (also called waves) different statistical information for a couple of variables.

In the first table, seen below, we have compared for each wave ID, the numbers and percentage of males, and of ethnicity category.

2.1 General statsitical overview per wave for sex and ethnicity variables

Table 1: Sex and ethnicity description of NHANES participants by wave

Wave ID	**2015** N = 10175	**2016** N = 9971	**2017** N = 9254
Male	5,003 (49%)	4,892 (49%)	4,557 (49%)
Ethnicity			
White	3,674 (36%)	3,066 (31%)	3,150 (34%)
Black	2,267 (22%)	2,129 (21%)	2,115 (23%)
Mexican	1,730 (17%)	1,921 (19%)	1,367 (15%)
Asian	1,074 (11%)	1,042 (10%)	1,168 (13%)
Other Hispanic	960 (9.4%)	1,308 (13%)	820 (8.9%)
Other	470 (4.6%)	505 (5.1%)	634 (6.9%)

[1] n (%)

Using the table above, we first observe a similar percentage of males for each wave, despite decreasing overall sample size of each wave. In addition to this, there seems to also be a general similarity of ethnicity proportion between waves.

Considering that all three waves are relatively similar in sex proportions and in ethnicity diversity proportions, we will follow with a statistical description of the average Systolic Blood Pressure (SBP) for each wave.

2.2 Average SBP statistical summary per wave

Below we find Table 2, presenting a summary of the average Systolic Blood Pressure per wave.

Table 2: Average systolic blood pressure by wave

Wave ID	**2015** N = 10175	**2016** N = 9971	**2017** N = 9254
Average Systolic Blood Pressure (mmHg)			
Min	64.7	74.0	72.7
Max	228.7	231.3	234.0
Median	115.3	117.3	118.0
Mean	118.3	120.4	121.7
SD	18.1	18.4	20.3
Unknown	2,647	2,608	2,541

Note: SD = standard deviation.

The table above allows us to observe the evolution of the average SBP throughout time for the general population of the United-States.

It is interesting to notice that, despite decrease of overall sample size and of missing values (see Unknown), there seems to be a **general increase** in the statistics of the average SBP. Indeed, we notice this by comparing values of each row for each survey cycle. In the span of 3 survey cycle (2014-2018):

- Min: **augmented** from 64.7 to 72.7 **(+8)**
- Max: **augmented** from 228.7 to 234.0 **(+5.3)**
- Median: **augmented** from 115.3 to 118.0 **(+2.7)**
- Mean: **augmented** from 118.3 to 121.7 **(+3.4)**
- Standard deviation (SD): **augmented** from 18.1 to 20.3 **(+2.2)**

Though this simple observation of data statistics is not complete enough to confirm this observed trend, let alone explain it, it is a first step to point out potential research that could further the understanding of this data.

End

References

Centers for Disease Control and Prevention (CDC), National Center for Health Statistics (NCHS). 2014. *National Health and Nutrition Examination Survey Data, 2013–2014.* U.S. Department of Health; Human Services, Centers for Disease Control; Prevention. https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx.

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